

Danté M. Herrera

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Awards

LANL SPOT Awards (2023, 2024)
LANL LAAP Award (2024)
1st Place – TAMU Student Research Week (Undergraduate Engineering, 2021)
National Hispanic Recognition Scholar
Dean's List (Multiple Terms)

Skills

Software

Python (advanced)
LabVIEW (advanced)
MATLAB / Simulink
C/C++ (microcontrollers)
SolidWorks, NX
HTML/CSS/JS, PHP, SQL
MS Office / LaTeX

Engineering

Large-scale DAQ systems
Sensor integration
Embedded controls
Signal conditioning
Automation & controls
Vacuum & gas systems

Collaboration

Technical communication
Team leadership
Experiment support
Mentoring

Coursework

Physics

Classical Mechanics I–II
Electricity & Magnetism I–II
Quantum Physics
Thermo & Stat Mech
Advanced Astrophysical Research
Advanced Biophysics Lab
Extragalactic Astronomy
Organic & Physical Chemistry

Aerospace / Applied

Computational Fluid Dynamics
Numerical Simulation
Heat Transfer & Viscous Flows
Active Controls for Aerospace Vehicles
Aerothermo & Propulsion Systems
High-Speed Aerodynamics
Calculus I–III
Differential Equations

Education

Arizona State University

B.S. Biophysics, *Summa Cum Laude*

2023–2025

GPA: 4.00

Texas A&M University

B.S. Aerospace Engineering (Honors), Astrophysics Minor
National Hispanic Recognition Scholar; GRE: 163V / 166Q

2018–2022

GPA: 3.79

Research & Engineering Experience

Los Alamos National Laboratory

2022–Present

R&D Engineer II – Gas Transfer Systems - Los Alamos, NM

- Lead developer for next-generation Gas Transfer Operation Station (GaTOS), integrating LabVIEW, embedded controllers, custom diagnostics hardware, and high-reliability monitoring for mission-critical operations.
- Developed the Valve Surveillance Test Stand (VSTS), a PXIe-based DAQ platform integrating accelerometers, analog/serial pressure transducers, thermocouples, RTDs, vacuum gauges, and custom diagnostic channels in support of critical NNSA hardware surveillance program.
- Built Python automation pipelines for calibration, data reduction, fault detection, and system verification/validation workflows.
- Supported major WETF plutonium and tritium experiments by designing DAQ packages, diagnostics integration, and automated control logic for high-consequence tests.
- Modernized legacy DAQ hardware by developing modular signal conditioning boards, improving noise performance and enabling the use of modern sensors.
- Collaborated with diagnostics, physics modeling, mechanical design, and operations teams to execute experimental campaigns.
- Trained new engineers and students on DAQ systems, LabVIEW / Python design practices, and experiment instrumentation workflows.

SpaceX

2021

Starship Launch Intern – Orbital Launch Pad Fluids - Boca Chica, TX

- Developed python based software tools for designing and monitoring fluid systems with real-time (stage-0) launch pad sensor data.
- Designed, installed, and operated cryogenic and pneumatic transfer systems supporting Starship integrated testing.
- Coordinated daily with fabrication, welding, and fluids teams to accelerate build-up and readiness of Orbital Launch Pad infrastructure.

TAMU Sounding Rocketry Team

2019–2021

Propulsion Team – Hybrid Engine Development
College Station, TX

- Designed hybrid engine combustion chamber, fuel grain geometry, and oxidizer routing using SolidWorks.
- Developed a small-scale R&D hybrid rocket motor enabling additive and grain-geometry testing with 15+ successful firings.
- Designed, built, and characterized a full-scale hybrid sounding rocket engine producing > 900 lbf average thrust.

Texas A&M Aerospace Engineering Dept.

2018–2021

Undergraduate Researcher – Hybrid Propulsion - College Station, TX

- Developed an HTPB/N₂O regression-rate model using heat transfer, mass flow, and convection models.
- Designed high temperature combustion-chamber temperature measurement and calibration techniques.
- Awarded **1st Place**, undergraduate general engineering poster section, TAMU Student Research Week 2021.